PHY 130A
Lecture Notes R:
Special Relativity and Relativistic Kinematics

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1 Vectors and Change of Basis

One of the first concepts encountered in introductory physics is the vector as a geometric object with both a magnitude and a direction. In more advanced subjects, like quantum mechanics, we introduce the abstract definition of a vector space V over a scalar field F (e.g. the real or complex numbers). The vector space is closed under vector addition and scalar multiplication, and has the following defining properties:

(1) $v + w = v + w$ for all $v, w \in V$.
(2) $v + (w + x) = (v + w) + x$ for all $v, w, x \in V$.
(3) There exists $0 \in V$ such that $v + 0 = v$ for all $v \in V$.
(4) For all $v \in V$ there exists $-v \in V$ such that $v + (-v) = 0$.
(5) $c(v + w) = cv + cw$ for all $v, w \in V$ and $c \in F$.
(6) $1v = v$ for all $v \in V$ (note: $1 \in F$).
(7) $(a + b)v = av + bv$ for all $v \in V$ and $a, b \in F$.
(8) $a(bv) = (ab)v$ for all $v \in V$ and $a, b \in F$.

As you will show in the exercise below, additional basic properties are deducible from these axioms:

(9)	$0 \in V$ is unique	(Show $0' + v = v \implies 0' = 0$)
(10)	$0v = 0$ for all $v \in V$	(Use $0 + 0 = 0$ for scalars)
(11)	$-v \in V$ is unique	(Show that $v + (-v') = 0 \implies -v = -v'$)
(12)	$(-1)v = -v$ for all $v \in V$	(Use $1 + (-1) = 0$ for scalars)

Note that we use the same symbol (0) for both the scalar zero and the zero vector; you must infer which one it refers to from context.

Exercise R.1: Deduce properties (9)-(12) from axioms (1)-(8) and known properties of the scalar field F. $\square$

The vector space $\mathbb{R}^n$ was the inspiration for the abstract vector space, but there are many more examples: complex numbers as vectors over the field of complex numbers, $n \times m$ real matrices over the scalar field of real numbers with ordinary matrix addition and scalar multiplication, real-valued periodic functions $f(x)$ of a real variable with period T over the scalar field of real numbers with ordinary addition and mulitiplication.

Exercise R.2: The set V is the rational numbers $\mathbb{R}$ and the field F is the real numbers, with ordinary addition and multiplication. Show that V is not a vector space. $\square$

Exercise R.3: The set V is the empty set. Is V a vector space? $\square$

Exercise R.4: For a vector space V scalar multiplication is a null operation, that is $av = v$ for all $a \in F$ and $v \in V$. Deduce all of the vectors in the vector space V. $\square$

Exercise R.5: (*) The set V is the positive reals, with all reals as the scalar field F. For clarity in what follows, we'll use $\oplus$ for vector addition and $\odot$ for scalar multiplication. Vector addition is defined as multiplication of the real numbers, which we write as:

$$v \oplus w = v\,w \qquad v, w \in V$$

for $w, v \in V$. Scalar multiplication is exponentiation, which we write as:

$$c \odot v = v^c \qquad v \in V, c \in F$$

Show that V is a vector space. What is the real number corresponding to the vector 0? $\square$

Exercise R.6: (*) For the previous exercise, verify explicitly that derived properties (10) and (12) hold for this vector space. $\square$

A basis $\mathcal{B}$ for a vector space V is an ordered set of vectors $\mathcal{B} = \{e_1, e_2, \cdots, e_m\}$ with the following properties:

- They are linearly independent, that is:

  $$c^1 e_1 + c^2 e_2 + \cdots + c^n e_n = 0$$

  only if $c^i = 0$ for all i.

- They span the space, that is for every $v \in V$ there exists a set of scalars $\{v^i \in F\}$ such that:

  $$v = \sum_i v^i e_i$$

  .

The scalar quantities v^i are called the components of the vector, and they depend on the basis $\mathcal{B}$ chosen. This is crucial point to understand, so it bears repeating: **a vector v exists independently of any basis, but its components v^i depend on the basis chosen $\mathcal{B} = \{e_i\}$.** Also, take care that e_i is not a scalar, it is a vector, in particular, it is this i-th basis vector. Don't let the index fool you into thinking e_i is a scalar like v^i.

Exercise R.7: (*) Show that the components v^i of a vector v for basis $\mathcal{B}$ are unique. (This is essential for many of the results that follow.) $\square$

Throughout this course, we will use the **Einstein summation convention**, whereby there is an implicit sum over repeated dummy indices with one a superscript and the other a subscript, e.g.:

$$v = v^i e_i \equiv \sum_i v^i e_i$$

Only **after** a basis $\mathcal{B}$ is chosen, it is possible to represent a vector as a column vector:

$$[v]_{\mathcal{B}} = \begin{pmatrix} v^1 \\ v^2 \\ \vdots \\ v^n \end{pmatrix}$$

or a row vector:

$$[v]_{\mathcal{B}}^T = \begin{pmatrix} v^1 & v^2 & \cdots & v^n \end{pmatrix}$$

.

Exercise R.8: (*) We can think of the complex numbers as 2-D vector space with the real numbers as scalars. Show that $e_1 = 1$ and $e_2 = i$ are a basis $\mathcal{B}$ for this vector space. For $v = a + bi$, write the column vector $[v]_{\mathcal{B}}$. This is why we are able to visualize the complex plane as a 2-D space. □

Exercise R.9: (*) The Pauli matrices:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

are a basis for the vector space of traceless 2×2 Hermitian matrices. Write the 2×2 matrix (an element of this vector space):

$$v = \begin{pmatrix} z & x - iy \\ x + iy & -z \end{pmatrix}$$

as a column vector $[v]_{\mathcal{B}}$ with basis $\mathcal{B} = \{\sigma_1, \sigma_2, \sigma_3\}$ □

Suppose we have two bases $\mathcal{B}$ and $\mathcal{B}'$. We have:

$$\mathcal{B} = \{e_i\}$$

as before, but for the primed basis we will write:

$$\mathcal{B}' = \{e_{i'}\}.$$

Notice that it is the index i that is primed in order to indicate that it is an index in the $\mathcal{B}'$ basis. The quantity e is not primed. We'll justify this convention shortly.

Since each e_i is a vector in V, and $\{e_{i'}\}$ is a basis, there must (by definition of basis, as above) exist scalar values $A^i_{j'}$ such that:

$$e_i = A_i^{j'} e_{j'}$$

(Remember there is an implicit sum across j' here.) Similarly, since each $e_{i'}$ is a vector in V, and $\{e_i\}$ is a basis, there must be some scalar values $A^j_{i'}$ such that:

$$e_{i'} = A^j_{i'} e_j$$

Remember, the location of the prime matters, so $A^i_{j'}$ refers to something altogether different from $A^j_{i'}$. But they are related:

$$e_i = A^{j'}_i e_{j'} = A^{j'}_i A^k_{j'} e_k$$

which requires (recall that the components of a basis are unique) that:

$$A^{j'}_i A^k_{j'} = \delta^k_i$$

and similarly:

$$A^i_{j'} A^{j'}_k = \delta^i_k$$

where

$$\delta^i_j = \begin{cases} 1 & i = j \\ 0 & \text{otherwise} \end{cases}$$

Any vector v can be expressed in terms of either basis $\mathcal{B}$ or basis $\mathcal{B}'$, which we write as:

$$v = v^i e_i = v^{j'} e_{j'}$$

but we also have:

$$e_{j'} = A^i_{j'} e_i \tag{1}$$

so that:

$$v^i e_i = v^{j'} e_{j'} = v^{j'} \left(A^i_{j'} e_i\right) = \left(A^i_{j'} v^{j'}\right) e_i$$

In a previous exercise you showed that the components of a vector within a basis (such as $\{e_i\}$ here) are unique. Therefore, we can conclude that for all i:

$$v^i = A^i_{j'} v^{j'}$$

In a similar fashion, we can find that:

$$v^{j'} = A^{j'}_i v^i$$

Compare this transformation rule to Equation 1, we see that the components v^i transform via $A^{j'}_i$, whereas the basis vectors e_i transform via $A^i_{j'}$. That is, they transform in a contrary fashion. **We call the vector components v^i contravariant, and we indicate this with an upper index. We call the basis vector e_i covariant, and we indicate this with the lower index.**

	index location	quantity	transformation	inverse transformation
v^i	upper index	contravariant component	$v^{i'} = A^{i'}_j v^j$	$v^i = A^i_{j'} v^{j'}$
e_i	lower index	covariant basis vector	$e_{i'} = A^j_{i'} e_j$	$e_i = A^{j'}_i e_{j'}$

Exercise R.10: Our first definition of a vector is a geometric object with a magnitude and direction. Suppose we scale each basis vector by a scalar s, to obtain a new basis. We must scale the components from the original basis by some scalar value in order to obtain new components for the new basis which preserve the magnitude of the vector. What is that scalar value? Does the "contra" in contravariant components make sense in this context? □

Exercise R.11: Consider the previous exercise, but instead of scaling, we rotate the original basis vectors by an angle θ about some axis w to obtain a new basis. Now we must rotate the components from the original basis about the axis w by some angle, in order to obtain new components for the new basis which preserve the direction of the vector. What is that angle? Does the "contra" in contravariant components make sense in this context? □

Exercise R.12: (*) Consider the vector space $\mathbb{R}^2$ as a vector space of the real numbers with the usual addition and scalar multiplication. Suppose that in the standard basis

$$\mathcal{B} = \{(1,0),(0,1)\}$$

the components of a vector v are:

$$[v]_{\mathcal{B}} = \begin{pmatrix} a \\ b \end{pmatrix}$$

For the alternative basis:

$$\mathcal{B}' = \{(1,1),(1,-1)\}$$

write the components of vector v as a column matric:

$$[v]_{\mathcal{B}'} = \begin{pmatrix} ? \\ ? \end{pmatrix}$$

2 Dual Vectors

For a vector space V over a scalar field F, we define a dual vector f as an F-valued linear function on V:

$$f : V \to F$$

That is, if f is a dual vector, then for any $v \in V$ we have $f(v) \in F$. The linearity means that for any $v, w \in V$ and $c \in F$ we have also:

$$f(v + c\,w) = f(v) + c\,f(w)$$

The set of all dual vectors to a vector space V is called the **dual space of** V, and it is denoted as V^*. As you will verify in the exercise below, the dual space is itself a vector space.

Exercise R.13: (**) For a vector space V over scaler field F, show that its dual space (V^*) is itself a vector space. Vector addition (addition of dual vectors) and scalar multiplication are defined such that:

$$(f + cg)(v) = f(v) + cg(v) \qquad\qquad f, g \in V^*,\ v \in V,\ c \in F$$

□

Assume that $\mathcal{B} = \{e_i\}$ is a basis for the vector space V. The action of any dual vector $f \in V^*$ on any vector $v \in V$ is completely determined by the action of f on the basis vectors e_i. For each i, we define a scalar quantity

$$f_i \equiv f(e_i) \qquad (f_i \in F)$$

and we see that:

$$f(v) = f(v^i e_i) = v^i f(e_i) = v^i f_i \tag{2}$$

which only depends on f via the scalar quantities f_i.

For each i, we also define a dual vector $e^i \in V^*$ by:

$$e^i(v) = v^i$$

We say that e^i picks off the ith component of v. If follows that

$$e^i(e_j) = \delta^i_j$$

because e_j is a vector in V whose components are all 0, except for the jth component, which is 1. You will show in the exercises below that the set of dual vectors $\{e^i\}$ are a basis for V^*, which we call the **dual basis** and denote as $\mathcal{B}^*$ You will also show that the scalar quanties f_i are in fact the components of f in the basis $\mathcal{B}$, so we have:

$$f = f_i e^i$$

which shows a dual vector f expanded in terms of its components f_i and basis vectors e^i.

The dual basis $\mathcal{B}^*$ depends entirely on the basis $\mathcal{B}$ of V. As $\mathcal{B}^*$ has one basis vector defined for each basis vector in $\mathcal{B}$, we see that **if V is finite dimensional, then it's dual space V^* has the same dimension.**

Exercise R.14: Verify from the definition of e^i that $e^i(v+cw) = e^i(v)+ce^i(w)$ so that e^i is indeed a dual vector. □

Exercise R.15: Show that

$$f = f_i e^i$$

is a valid expansion for f, by verifying that

$$f(v) = v^i f_i$$

as required by Equation 2. This shows that $\{e^i\}$ spans the dual space. □

Exercise R.16: (*) Show that if: $a_i e^i = 0$ then, for all i $a_i = 0$ This shows that $\{e^i\}$ is linearly independent. Hint: define the dual vector $a = a_i e^i$ and evaluate $a(e_i)$ for each i. □

We showed previously that the components of a vector v tranform in a covariant fashion, so we call the v^i the covariant components. Suppose that our basis transforms according to:

$$e_{i'} = A^j_{i'} e_j$$

then we see that:

$$f_{i'} = f(e_{i'}) = f(A^j_{i'} e_j) = A^j_{i'} f(e_j) = A^j_{i'} f_j$$

so the components of the dual vector f_i transform covariantly. Notice how nicely our notation tracks covariance and contravariance. As long as we keep indices in the same position on both side of equations, avoiding constructions like:

$$f^i = f(e_i) \qquad (wrong!)$$

then linearity ensures the covariance and contravariance of quantities can be read off from the location of the index. (This is an important practical matter, so take the time to understand it.)

To recap, we can expand a vector $v \in V$ in terms of any basis $\mathcal{B} = \{e_i\}$ and its scalar components $v^i \in F$:

$$v = v^i e_i.$$

We can likewise expand a dual vector $f \in V^*$ in terms of the dual basis $\mathcal{B}^*$ and its scalar components $f_i \in F$:

$$f = f_i e^i$$

Notice that the location of the indices in dual vectors is opposite that of vectors, reflecting that dual vectors have covariant components and vectors have contravariant components. We can think of the scalar components v^i and f_i in two complementary ways. First, v^i and f_i are the unique set of scalars which scale the basis vectors of $\mathcal{B}$ and $\mathcal{B}^*$ to the vectors v and f. Secondly, the components can both be obtained by evaluating a dual vector:

$$v^i = e^i(v)$$

and

$$f_i = f(e_i)$$

In the vector space $\mathbb{R}^3$ we are familiar with obtaining the components of a vector by taking dot products. In a Hilbert space, we do so with an inner product. But here, we have not defined an inner product. We see that dual vectors are a generalization of the dot product idea.

Exercise R.17: Using the definition, show that e^i transforms contravariantly, as expected for a quantity with upper indices.

3 Introduction to Tensors

A traditional physics definition of a tensor is a generalization of scalars and vectors, to include quantities with muliple indices which transform in a specific way when changing the basis. This definition is perfectly reasonable, but it is not particularly insightful, and things (like the transformation rules) seem to come from nowhere. We are going to use a more modern definition of a tensor, which is equivalent, but clarifies the situation considerably.

Let V be vector space and V^* its dual space. We define an (r, s) tensor as an F-valued multilinear function of r vectors and s dual vectors:

$$T : \underbrace{V \times V \times \cdots \times V}_{r \text{ times}} \times \underbrace{V^* \times V^* \times \cdots \times V^*}_{s \text{ times}} \to F$$

The scalar value returned by a tensor is written as:

$$T(v_1, v_2, \cdots, v_n, f_1, f_2, \cdots, f_s)$$

where $v_1, v_2, \cdots, v_r \in V$ and $f_1, f_2, \cdots, f_n \in V^*$ are **not** components of vectors or dual vectors but r, despite the indices, but r entire vectors and s entire dual vectors. We define addition and scalar multiplication in the obvious manner, as clarified by example below. This makes the set of all (r, s) tensors a vector space, which we call $\mathcal{T}_s^r(V)$.

The proliferation of indices makes talking about general tensors cumbersome, so let's work things out explicitly for a (1,1) tensor even though our conclusions apply to all tensors. We define tensor addition of two (1,1) tensors T and S by:

$$(T+S)(v,f) = T(v,f) + S(v,f)$$

with $v \in V$ and $f \in V^*$. We define scalar multiplication by:

$$(cT)(v,f) = c\,(T(v,f))$$

with $c \in F$. Multilinearity means that:

$$\begin{aligned} T(v+cw,f) &= T(v,f) + c\,T(w,f) \\ T(v,f+cg) &= T(v,f) + c\,T(v,g) \end{aligned}$$

for $w \in V$ and $g \in V^*$.

Under the traditional definition, we consider tensors to be a generalization of the concept of scalars and vectors, so that scalar and vectors are examples of tensors. Is that the case under the new definition? The only immediately obvious connection is that a $(0,1)$ tensor precisely reproduces the definition of a dual vector. It is also convenient to consider (0,0) tensors as scalars. That is, we can think of scalar as a scalar-valued function with zero vector and zero dual vector arguments.

It is a bit more challenging to connect vectors to tensors under the new definition. Formally, we say that vectors are isomorphic to the (1,0) tensors by mapping each vector $v \in V$ to a (1,0) tensor T_v

$$v \mapsto T_v$$

defined by:

$$T_v(f) \equiv f(v).$$

Less formally, we are going to play a little game, you and I. You only have an ordinary vector v, but you are going to convince me that you have a (0,1) tensor. So I give you $f \in V^*$ and, quite cleverly really, you put your vector into my dual vector, and give me back $f(v)$. This shows me that you have an F-valued function of V^*, as required for a (0,1) tensor. But is it linear? I give $g \in V^*$ and you give me $g(v)$. I give you $f + cg$ with $c \in F$ and you give me:

$$(f+cg)(v) = f(v) + cg(v)$$

which verifies that you have a linear function. So you have now convinced me that you have a $(0,1)$ tensor, even though you really only had a vector. One more interpretation: a $(0,1)$ tensor is the dual of a dual vector, and so a vector.

Next we'll consider a (2,0) tensor T. By definition T as an F-valued multilinear function of two vectors:

$$T : V \times V \to F$$

and the scalar value returned for $v, w \in V$ is written as

$$T(v,w).$$

A tensor is completely described by its action on any set of basis vectors. The components of T are defined as

$$T_{ij} \equiv T(e_i, e_j)$$

we can calculate

$$T(v, w) = T(v^i\, e_i, w^j\, e_j) = v^i T_{ij} w^j,$$

which shows we can calculate $T(v, w)$ from only the components T_{ij}.

We can clearly see the difference between a tensor and a matrix. Remember that vectors are geometric objects that exist independently of any particular basis. Only after a basis has been selected, we can refer to them by their components, and arrange their components into row or column vectors. In exactly the same way, a tensor exists independently of any basis. But once we define a basis $\mathcal{B}$ we can refer to a vector by it's components. We can arrange the components of a vector into a row vector (denoted $[v]_{\mathcal{B}}$) or a column vector (denoted $[v]_{\mathcal{B}}^T$). Likewise, we can arrange the compments of a (2,0) tensor into a matrix, denoted $[T]_{\mathcal{B}}$. We can calculate the action of T on vectors v and w via matrix multiplication:

$$T(v, w) = v^i\, T_{ij}\, w^j = [v]_{\mathcal{B}}^T\, [T]_{\mathcal{B}}\, [v]_{\mathcal{B}} = \begin{pmatrix} v^1\, v^2\, \cdots\, v^n \end{pmatrix} \begin{pmatrix} T_{11} & T_{12} & \cdots & T_{1n} \\ T_{21} & T_{22} & \cdots & T_{2n} \\ \vdots & \vdots & \ddots & \\ T_{n1} & T_{n2} & & T_{nn} \end{pmatrix} \begin{pmatrix} w^1 \\ w^2 \\ \vdots \\ w^n \end{pmatrix}$$

Another benefit of the modern definition is that we can derive the transformation law for the components of the tensor, instead of treating that as part of the definition.

Exercise R.18: (*) Suppose that a basis $\mathcal{B}$ is related to another basis $\mathcal{B}'$ by:

$$e_{i'} = A^j_{i'} e_j$$

and

$$e_i = A^{j'}_i e_{j'}$$

Use multilinearity to show that:

$$T_{i'j'} = A^k_{i'} A^l_{j'} T_{kl}$$

and:

$$T_{ij} = A^{k'}_i A^{l'}_j T_{k'l'}$$

□

Exercise R.19: For a vector space V over scaler field F, show that the set of all (2,0) tensors is a vector space.

Suppose we have a (1,1) tensor L so that $L : V \times V^* \to F$. For any vector v and basis $\mathcal{B}$, we can define a vector w by its components:

$$w^i = L(v, e^i) = L(v^j e_j, e^i) = L(e_j, e^i) v^j$$

and so with components of L defined as:

$$L(e_j, e^i) \equiv L^i_j$$

we have

$$w^i = L^i_j v^j$$

We can thus interpret as a (1,1) tensor as a linear operator that maps $V \to V$.

4 Non-degenerate Hermitian Forms and Metric Duals

In quantum mechanics, we saw that a Hilbert space H is a vector space with an inner product defined. The inner product is positive definite:

$$\langle v|v\rangle > 0$$

for $v \in H$ and $v \neq 0$. It turns out that this requirement is too restrictive for our purposes (special relativity). We will need a slightly more general concept than the inner product. We will use the bar symbol to denote complex conjugation, so $\bar{z}$ denotes the complex conjugate of z for $z \in \mathbb{C}$. If the bar symbol is applied to a real number, it has no effect.

A non-degenerate Hermitian form on a vector space V is an F-valued function, denoted $(\cdot|\cdot)$, which assigns to any ordered pair of vectors v and w the scalar value denoted $(v|w)$, and has the following properties:

(1) Linear in the second slot:
$(v|w + c\,x) = (v|w) + c\,(v|x)$ for all $v, w, x \in V$ and $c \in F$.

(2) Conjugate symmetry:
$(v|w) = \overline{(w|v)}$ for all $v, w \in V$.

(3) Non-degenerate:
For $v \in V$ and $v \neq 0$ there exists $w \in V$ with $(v|w) \neq 0$.

Now hopefully properties (1) and (2) are quite familiar properties of the inner products you already encountered in quantum mechanics, but property (3) is new. Notice, however, that property (3) is automatically satisfied for a Hilbert space, by choosing $w = v$ and appealing to positive definiteness. So conceptually, a non-degenerate form is an inner product but for the requirement of positive definiteness, which is loosened to property (3) instead.

Exercise R.20: Show that for a non-degenerate Hermitian form $(\cdot|\cdot)$ is conjugate-linear in the first slot:

$$(v + c\,x|w) = (v|w) \;+\; \bar{c}\,(x|w)$$

for all $v, w, x \in V$ and $c \in F$ $\square$

Given a vector $v \in V$ with non-degenerate Hermitian form $(\cdot|\cdot)$ we can define a dual vector $\tilde{v}$ by:

$$\tilde{v}(w) = (v|w)$$

for all $w \in V$. We call the dual vector $\tilde{v}$ the metric dual of v. We sometimes write:

$$\widetilde{v} = (v|\cdot)$$

.

Now consider the map $L : V \to V^*$ that sends a vector to it's metric dual:

$$L(v) \equiv \tilde{v}$$

In what follows, we will show that L establishes a conjugate-linear bijection from V to V^*. That is:

- L is conjugate linear.
- L is one-to-one.
- L is onto.

This established a conjugate-linear isomorphism sending each vector to it's metric dual. In laymen's terms, this shows that the dual space is essentially a copy of the vector space, where each vector corresponds uniquely to a dual vector, and the structure of vector addition and scalar multiplication is preserved (with complex scalars conjugated.

Exercise R.21: Show that L is conjugate linear, that is for $v, x \in V$ and $c \in F$ then

$$L(v + c\,x) = L(v) + \bar{c}\,L(x)$$

Hint: $(L(v + c\,x))(w) = (v + c\,x|w)$ $\square$

Exercise R.22: Show that L sends zero to zero. That is, show $L(0) = 0$. $\square$

Exercise R.23: Show that the only vector L sends to zero is zero, that is:

$$L(w) = 0 \implies w = 0$$

$\square$

Exercise R.24: (*) Show that L is a one-to-one map of $V \to V^*$, that is, show that:

$$L(w) = L(v) \implies w = v$$

Hint: use non-degeneracy, i.e. if for all $x \in V$

$$(v|x) = 0$$

then

$$v = 0$$

$\square$

Let $\mathcal{B} = \{e_i\}$ be a basis for V. The set of dual vectors $\{L(e_i)\}$ are (as we will show) a basis for V^*. First, we show that $\{L(e_i)\}$ is linearly independent. Assume there is some set of scalars a^i such that:

$$a^i L(e_i) = 0$$

Then define a vector $a \in V$ by

$$a \equiv a^i e_i$$

We have:

$$L(a) = L(a^i e_i) = a^i L(e_i) = 0$$

This implies (see previous exercises) that $a = 0$, and so all of it's components are also 0:

$$a^i = 0$$

This shows that $\{L(e_i)\}$ is linearly independent. Next, we show that $\{L(e_i)\}$ spans the space. For a finite-dimensional vector space V, recall that the dual space V^* has the same dimension as V. We showed that L is one-to-one, and so $\{L(e_i)\}$ has the same number of elements as $\mathcal{B}$, which is the dimension of V^*. As they are linearly independent, they therefore must span the space V^*. Therefore $\{L(e_i)\}$ is a basis for V^*

Exercise R.25: Given that $\{L(e_i)\}$ is a basis for V^*, show that the mapping L from V to V^* is onto. That is, show that for any $b \in V^*$, there is an $a \in V$ such that

$$L(a) = b$$

□

By now we have shown that the mapping L is both one-to-one and onto, which means that it is an invertible mapping. **We can move freely from a vector to its metric dual and back again with no ambiguity.**

For a Hilbert space H with inner product $\langle\cdot|\cdot\rangle$. In Dirac notation, a vector $\psi \in H$ is written as a ket $|\psi\rangle$. It's metric dual $L(\psi)$ is written as a bra $\langle\psi|$.

Exercise R.26: (*) For $\psi \in H$ and $c \in F$, show that

$$\langle c\psi| = \bar{c}\,\langle\psi|$$

□

Exercise R.27: (*) Consider the vector space $\mathbb{R}^2$ as a vector space of the real numbers with the usual addition and scalar multiplication. Take the vector dot product as an NDHF (now restricted to reals):

$$(a, b) = \vec{a} \cdot \vec{b}$$

Suppose that in the standard basis

$$\mathcal{B} = \{(1, 0), (0, 1)\} = \{\hat{x}, \hat{y}\}$$

the components of a vector v are:

$$[v]_{\mathcal{B}} = \begin{pmatrix} a \\ b \end{pmatrix}$$

What are the components of the metric dual $\tilde{v}$ in the dual basis $\mathcal{B}^*$, written as a column vector:

$$[\tilde{v}]_{\mathcal{B}^*} = \begin{pmatrix} ? \\ ? \end{pmatrix}$$

(This is the reason that you can do a whole lot of linear algebra never having to discuss dual vectors.)

Exercise R.28: For the previous problem, let L be the linear map that maps v to $\tilde{v}$, again using the usual dot-product. What are the components of the metric dual $\tilde{v}$ in the basis $\mathcal{L} = \{L(\hat{x}), L(\hat{y})\}$?

$$[\tilde{v}]_{\mathcal{L}} = \begin{pmatrix} ? \\ ? \end{pmatrix}$$

That completes the math prelimineraries we will need for our invariance-based treatment of special relativity.

5 Invariance of the Space-Time Interval

Special relativity, as formulated by Einstein, is based on two postulates:

(1) Physical laws are the same for all inertial frames
(2) The speed of light in free space has the same value c in all inertial frames

In what follows, we will only consider inertial frames, so we will drop the word "inertial". We will also only consider inertial frames with a relative velocity less than the speed of light. In the exercises, you will explore the paradoxes that follow if inertial frames exist with a relative velocity equal to or greater than the speed of light.

Suppose that in some frame F, light is emitted at time t_{A} and position $(x_{\mathrm{A}}, y_{\mathrm{A}}, z_{\mathrm{A}})$. The light travels through free space until some later time t_{B}, when the light is aborbed at position $(x_{\mathrm{B}}, y_{\mathrm{B}}, z_{\mathrm{B}})$. Because the speed of light in vacuum is the universal constant c we have:

$$c = \frac{\sqrt{(x_{\mathrm{B}} - x_{\mathrm{A}})^2 + (y_{\mathrm{B}} - y_{\mathrm{A}})^2 + (z_{\mathrm{B}} - z_{\mathrm{A}})^2}}{t_{\mathrm{B}} - t_{\mathrm{A}}}$$

or equivalently:

$$c^2(t_{\mathrm{B}} - t_{\mathrm{A}})^2 - (x_{\mathrm{B}} - x_{\mathrm{A}})^2 - (y_{\mathrm{B}} - y_{\mathrm{A}})^2 - (z_{\mathrm{B}} - z_{\mathrm{A}})^2 = 0$$

In any other inertial frame F', in which the light is emitted at time t'_{A} and position $(x'_{\mathrm{A}}, y'_{\mathrm{A}}, z'_{\mathrm{A}})$ and absorbed at time t'_{B} and position $(x'_{\mathrm{B}}, y'_{\mathrm{B}}, z'_{\mathrm{B}})$,

$$c^2(t'_{\mathrm{B}} - t'_{\mathrm{A}})^2 - (x'_{\mathrm{B}} - x'_{\mathrm{A}})^2 - (y'_{\mathrm{B}} - y'_{\mathrm{A}})^2 - (z'_{\mathrm{B}} - z'_{\mathrm{A}})^2 = 0$$

Quantities that have the same value in every reference frame are precious. So this result alone is enough motivation to define the space-time interval ΔS^2 between any two events A and B as:

$$\Delta S^2 \equiv c^2(t_{\mathrm{B}} - t_{\mathrm{A}})^2 - (x_{\mathrm{B}} - x_{\mathrm{A}})^2 - (y_{\mathrm{B}} - y_{\mathrm{A}})^2 - (z_{\mathrm{B}} - z_{\mathrm{A}})^2 \tag{3}$$

The space-time interval thus defined has several properties that are immediately evident:

(A) If $\Delta S^2 = 0$ in one frame, then it is zero in every frame.
(B) ΔS^2 is invariant with respect to rotations in space.
(C) ΔS^2 is invariant with respect to translations in space and time.

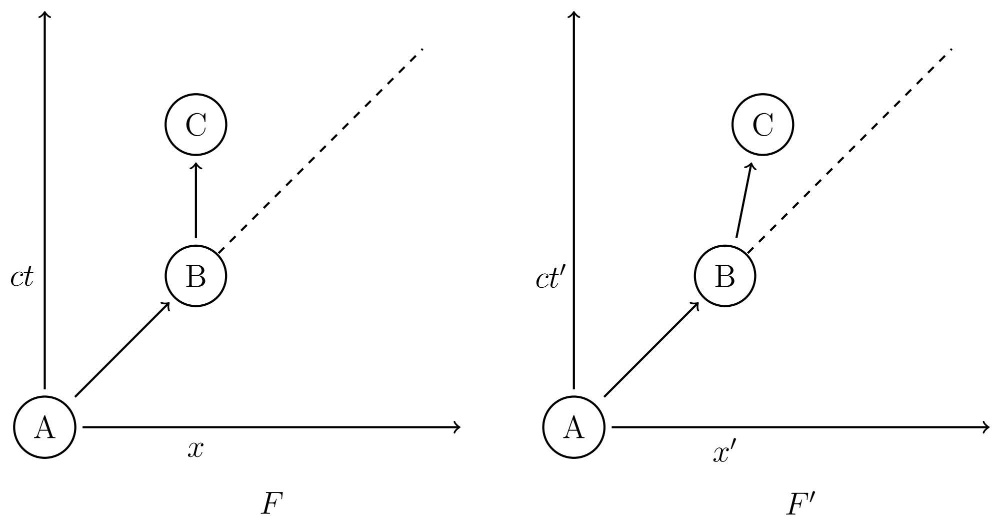


Figure 1: A light pulse is emitted at event A and absorbed and recorded at event B, as seen in two frames F and F'. The record of the event remains at rest in frame F.

Property (A) is evident from the calculation that motivated the definition. Property (B) follows because ΔS^2 only involves the spatial components via the usual Euclidean metric:

$$\Delta D^2 \equiv (x_{\mathrm{B}} - x_{\mathrm{A}})^2 + (y_{\mathrm{B}} - y_{\mathrm{A}})^2 + (z_{\mathrm{B}} - z_{\mathrm{A}})^2$$

which is invariant under rotations. Property (C) follows because ΔS^2 only involves differences in time and spatial components, e.g. $t_{\mathrm{B}} - t_{\mathrm{A}}$.

Next we'll show that the following additional property holds:

(D) The sign of ΔS^2 is invariant

Suppose (as shown in Fig. 1) there are two events, labeled A and C, and some frame F where $\Delta S^2 > 0$. In frame F, we emit a light pulse at Event A. The light pulse travels through vacuum until it reaches the location of Event C, where it is absorbed and recorded, at an event we label B. The record remains at rest until event C. Now consider how this looks in some other frame F' traveling at velocity $v < c$ wrt frame F. The light is emitted at Event A, and absorbed and recorded at Event B, but now the record travels at speed $v < c$ until Event C. Since $v < c$, the distance between Event A and Event C must therefore be less than $c(t'_{\mathrm{B}} - t'_{\mathrm{A}})$ so $\Delta S^2 < 0$.

We categorize space time intervals by the sign of the space-time interval as follows:

$$\begin{array}{lcl} \Delta S^2 = 0 & \Longleftrightarrow & \text{light-like} \\ \Delta S^2 < 0 & \Longleftrightarrow & \text{space-like} \\ \Delta S^2 > 0 & \Longleftrightarrow & \text{time-like} \end{array}$$

From property (A) we see that if the space-time interval between two events in one frame is light-like, it is light-like in all frames. We have also just shown that if the interval is space-like in one frame, it is space-like in all frames. It follows that if the interval is time-like in one frame, it is time-like in all frames (or else we would have a contradiction of one of the previous two results). Which demonstrates property (D).

Finally, we will show that ΔS^2 is invariant. Consider an infinitesimal space-time inverval calculated in frame F:

$$dS^2 = dt^2 - dx^2 - dy^2 - dz^2 \tag{4}$$

and the corresponding quantiry in frame F':

$$(dS^2)' = (dt')^2 - (dx')^2 - (dy')^2 - (dz')^2 \tag{5}$$

Now $(dS^2)'$, which is quadratic in differentials of the primed coordinates, must also be quadratic in differentials of the unprimed coordinates, because, as shown in the appendix, coordinate transformations are linear.

We can formally expand $(dS^2)'$ as a power series in dS^2:

$$dS^{2'} = a + bdS^2 + c(dS^2)^2 + \cdots$$

with coefficients $a, b, c, \cdots$ to be determined. From property (A) we see that $a = 0$. But $(dS^2)^2$ is quartic (power 4) in differentials of the coordinates, while $(dS^2)'$ is quadratic, and so higher order terms (c and beyond) vanish. This leads to:

$$dS^{2'} = bdS^2$$

with b still unknown. But b cannot depedent on the absolute position in space and time, because of property (C). It can depend on the relative motion between the F and F' frames, but not the direction, because of property (B). So it can only dependent on the relative speed of F and F'. But this is the same in both frames, so we must also have:

$$dS^2 = bdS^{2'}$$

and so:

$$dS^2 = b^2 dS^2$$

from which we conclude $b = \pm 1$. But $b = -1$ would violate property (D). So:

$$(dS^2)' = dS^2$$

and (by integration) ΔS^2 is invariant as well.

Thus we conclude with the main result of this section, **the space-time interval defined as:**

$$\Delta S^2 = \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2 \tag{6}$$

is frame invariant. Note that we have set $c = 1$ in writing Equation 6, as will be our convention from here on.

Exercise R.29: Consider Fig. 1 and suppose there is a frame F'' in which events A and B occur at the spatial origin. What is the speed of of frame F'' in frame F? Show that events A and B must occur simultaneously in frame F. Note that as a result, the spatial origin of frame F'' at $t'' = 0$ is located at both events A and B in the frame F. So therefore there is no invertable map of space-time points between frame F and F''. We conclude that it is not possible for an inertial frame to travel at the speed of light as viewed from any other inertial frame.

6 Four-Vectors and the Metric of Space-Time

In order to represent points in space-time, we use the space $\mathbb{R}^4$, which is a vector space with the standard basis $\mathcal{B}$ consisting of of the following vectors:

$$\begin{aligned}
e_0 &= (1,0,0,0)\\
e_1 &= (0,1,0,0)\\
e_2 &= (0,0,1,0)\\
e_3 &= (0,0,0,1)
\end{aligned}$$

We define a **position-space four-vector** (v) as a vector from the vector space $\mathbb{R}^4$, which, under the standard basis $\mathcal{B} = \{e_\mu\}$, has contravariant components v^μ matched to space-time coordinates from frame F as:

$$v^0 = t, \qquad v^1 = x, \qquad v^2 = y, \qquad v^3 = z. \tag{7}$$

It is conventional to use greek indices, which run from from 0 to 3, when referring to the components of four-vectors:

$$v = v^\mu e_\mu$$

Associated with the four-vector space, we define a non-degenerate Hermitian form:

$$(a|b) = a^0b^0 - a^1b^1 - a^2b^2 - a^3b^3 \tag{8}$$

Exercise R.30: Show that $(\cdot|\cdot)$ as defined above is a non-degenerate Hermitian form, by demonstrating that it has the three definining properties. $\square$

Exercise R.31: For position four-vectors $a, b \in V$ with space time interval $\Delta S^2(a,b)$ between them, show that:

$$(b-a|b-a) = \Delta S^2(a,b)$$

$\square$

Exercise R.32: Show that $(\cdot|\cdot)$ as defined above is **not** an inner product. $\square$

We can interpret our non-degenerate Hermitian form as multilinear map $g : V \times V \to R$. That is, as a (2,0) tensor:

$$g(a,b) \equiv (a|b)$$

As always, we can work out the components of the tensor in the standard basis $\mathcal{B}$ with

$$g_{\mu\nu} \equiv g(e_\mu, e_\nu) = (e_\mu|e_\nu) \tag{9}$$

So now we can write:

$$(a|b) = g(a,b) = g(a^\mu e_\mu, b^\nu e_\nu) = a^\mu b^\nu g(e_\mu, e_\nu) = a^\mu b^\nu g_{\mu\nu}$$

For this reason, we call g the **metric tensor** (recall that the $(\cdot|\cdot)$ was defined based on the space-time interval.)

In the exercises you will show that

$$g_{00} = 1, \qquad g_{11} = g_{22} = g_{33} = -1 \tag{10}$$

and all other elements are zero. Thus, the metrix tensor g can be represented by a matrix G as:

$$[g]_{\mathcal{B}} = G \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \tag{11}$$

Now, there is nothing special about the frame F and its standard basis $\mathcal{B}$. Therefore, we must reach the same conclusion about the matrix represenation of G in any other iniertial frame F' with standard basis $\mathcal{B}'$. Specifically, we must have:

$$[g]_{\mathcal{B}} = [g]_{\mathcal{B}'} = G$$

for all frames. The components of the metric tensor are the same in all intertial frames. Here, our primed index notation becomes just a little bit cumbersome, but we can write this as:

$$g_{\alpha'\beta'} = \delta^{\alpha}_{\alpha'}\delta^{\beta}_{\beta'}g_{\alpha\beta} \tag{12}$$

where the δ factors are just an artifact of our notation. In matrix notation, this reads:

$$G' = I\,G\,I = G$$

Exercise R.33: Verify Equations 10 and 11 by calculating the matrix components. □

Exercise R.34: Use Equation 12 to show that for any two position-space four vectors a and b, the quantity

$$(a|b) = a^{\mu}b_{\mu}$$

is frame invariant. □

Notice that g is its own inverse, so defining:

$$g^{\mu\nu} \equiv g_{\mu\nu}$$

we have

$$g^{\mu\alpha}g_{\alpha\nu} = \delta^{\mu}_{\nu}$$

With a Non-Degenerate Hermitian Form $(\cdot|\cdot)$ defined in Equation 8, then, as shown in Section 4, to each vector $v \in V$ we can associate a metric dual

$$\tilde{v} \equiv (v|\cdot)$$

which is shorthand for defining its action on any vector $w \in V$

$$\tilde{v}(w) = (v|w)$$

We can also fully describe $\tilde{v}$ by its components with respect to the basis vectors $\mathcal{B} = \{e_\mu\}$, because:

$$\tilde{v}(w) = \tilde{v}(w^\mu e_\mu) = w^\mu \tilde{v}(e_\mu) = w^\mu \tilde{v}_\mu$$

where the components of $\tilde{v}$ are:

$$\tilde{v}_\mu \equiv \tilde{v}(e_\mu) = (v|e_\mu) = g(v, e_\mu) = g(v^\nu e_\nu, e_\mu) = v^\nu g(e_\nu, e_\mu)$$

leading to the very useful result:

$$\tilde{v}_\mu = v^\nu g_{\nu\mu} = g_{\mu\nu} v^\nu \tag{13}$$

So we see that we can use the metric tensor g to lower indices.

Exercise R.35: (*) Show that:

$$v^\mu = g^{\mu\nu}\, \tilde{v}_\nu$$

so that we can use the metric tensor to raise indices as well. Hint: $g^{\mu\alpha}\, g_{\alpha\nu} = \delta^\mu_\nu$.

Notation: It is common practice to ommit the tilde from the dual of a four-vector $\tilde{v}_\mu$ as it can be inferred from the lowered index:

$$v_\mu \equiv \tilde{v}_\mu = g_{\mu\nu}\, v^\nu$$

Notice how our tensor definition applies here. The metric tensor is a mapping $g : V \times V \to F$. With one vector given, the metric tensor thus provides a map from $V \to F$, which is a dual vector. So we can interpret a (2,0) tensor has a linear operator from $V \to V^*$, exactly as seen here with $g_{\mu\nu}$. Similary, $g^{\mu\nu}$ can be seen as the components of a (0,2) tensor and as a linear operator from $V^* \to V$.

Exercise R.36: In matrices, the lowering of an index reads:

$$[\tilde{v}]_{\mathcal{B}^*} = G\,[v]_{\mathcal{B}}$$

Consider the position four-vector with components in the standard basis $\mathcal{B}$:

$$[v]_{\mathcal{B}} = \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix}$$

Using matrices, show that the components of its metric dual in the dual basis are

$$[\tilde{v}]_{\mathcal{B}^*} = \begin{pmatrix} t \\ -x \\ -y \\ -z \end{pmatrix}$$

7 Coordinate Transformations

Suppose that in the vector space $\mathbb{R}^4$ for position-space four-vectors, the standard basis vectors $\mathcal{B}$ of frame F are related to the standard basis vector $\mathcal{B}'$ of frame F' by a coordinate transformation:

$$e_{\mu'} = \Lambda^\nu_{\ \mu'}\, e_\nu.$$

We showed before that there is an inverse transformation:

$$e_\mu = \Lambda^{\nu'}_{\mu}\, e_{\nu'}$$

with:

$$\Lambda^{\beta}_{\eta'}\Lambda^{\eta'}_{\alpha} = \delta^{\beta}_{\alpha}$$

As shown in Section 1, the contravariant components of a position-space four-vector v will therefore transform as:

$$v^{\mu'} = \Lambda^{\mu'}_{\nu} v^{\nu}$$

Any vector in $\mathbb{R}^4$ whose contravariant components transform in the same manner as a position-space four-vector is called a **four-vector**.

We might be tempted to consider the real numbers $\Lambda^{\nu'}_{\mu}$ as components of a (1,1) tensor. However, notice that one set of indices refers to the $\mathcal{B}$ basis and another the $\mathcal{B}'$. Therefore, it is best to leave tensors aside and interpret $\Lambda^{\mu'}_{\nu}$ as the components of a matrix, labeled by the two sets of basis vectors which it transforms between:

$$[\Lambda]^{\mathcal{B}'}_{\mathcal{B}} \equiv \begin{pmatrix} \Lambda^{0'}_{0} & \Lambda^{0'}_{1} & \Lambda^{0'}_{2} & \Lambda^{0'}_{3} \\ \Lambda^{1'}_{0} & \Lambda^{1'}_{1} & \Lambda^{1'}_{2} & \Lambda^{1'}_{3} \\ \Lambda^{2'}_{0} & \Lambda^{2'}_{1} & \Lambda^{2'}_{2} & \Lambda^{2'}_{3} \\ \Lambda^{3'}_{0} & \Lambda^{3'}_{1} & \Lambda^{3'}_{2} & \Lambda^{3'}_{3} \end{pmatrix}$$

where we have left the primes in place, even on numerical values, because in our notation the location of the primed indice (i.e. $\Lambda^{1'}_{2}$ means $\Lambda^{\mu'}_{\nu}$ for $\mu' = 1$ and $\nu = 2$.) Notice that we have choosen to associate the **upper** index with the row of the matrix, which is simply a matter of choice, but one which we must apply consistently. In matrix notation, we have:

$$[v]_{\mathcal{B}'} \;=\; [\Lambda]^{\mathcal{B}'}_{\mathcal{B}}\; [v]_{\mathcal{B}}$$

which for for a position-space four vector is:

$$\begin{pmatrix} t' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \Lambda^{0'}_{0} & \Lambda^{0'}_{1} & \Lambda^{0'}_{2} & \Lambda^{0'}_{3} \\ \Lambda^{1'}_{0} & \Lambda^{1'}_{1} & \Lambda^{1'}_{2} & \Lambda^{1'}_{3} \\ \Lambda^{2'}_{0} & \Lambda^{2'}_{1} & \Lambda^{2'}_{2} & \Lambda^{2'}_{3} \\ \Lambda^{3'}_{0} & \Lambda^{3'}_{1} & \Lambda^{3'}_{2} & \Lambda^{3'}_{3} \end{pmatrix} \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix}$$

In similar fashion, we can consider $\Lambda^{\nu}_{\mu'}$ as the components of a matrix, labeled by the two sets of basis vectors it connects:

$$[\Lambda]^{\mathcal{B}}_{\mathcal{B}'} \equiv \begin{pmatrix} \Lambda^{0}_{0'} & \Lambda^{0}_{1'} & \Lambda^{0}_{2'} & \Lambda^{0}_{3'} \\ \Lambda^{1}_{0'} & \Lambda^{1}_{1'} & \Lambda^{1}_{2'} & \Lambda^{1}_{3'} \\ \Lambda^{2}_{0'} & \Lambda^{2}_{1'} & \Lambda^{2}_{2'} & \Lambda^{2}_{3'} \\ \Lambda^{3}_{0'} & \Lambda^{3}_{1'} & \Lambda^{3}_{2'} & \Lambda^{3}_{3'} \end{pmatrix}$$

These matrices are related by:

$$\Lambda^{\beta}_{\eta'}\Lambda^{\eta'}_{\alpha} = \delta^{\beta}_{\alpha}$$

which in terms of the above matrices reads:

$$[\Lambda]^{\mathcal{B}}_{\mathcal{B}'}\; [\Lambda]^{\mathcal{B}'}_{\mathcal{B}} = I$$

which is as one would hope.

Notation: When there is little chance of confusion, we drop the explicit labeling by basis vectors, and speak of the transformation Λ, with matrix representation

$$\Lambda \equiv [\Lambda]_{\mathcal{B}}^{\mathcal{B}'}$$

which would implicitly refer to the transformation applicable to the contravariant components when moving from frame F to F' (or, equivalently, to the covariant components when moving from frame F' to F). The inverse transformation is:

$$\Lambda^{-1} = [\Lambda]_{\mathcal{B}'}^{\mathcal{B}}$$

which is the transformation applicable to the contravariant components when moving from frame F' to F (or, equivalently, to the covariant components when moving from frame F to F').

Note that Giffiths uses the notation:

$$(v^\mu)' = \Lambda^\mu_\nu \, v^\nu$$

so

$$\Lambda^{\mu'}_\nu \longrightarrow \Lambda^\mu_\nu \quad \text{(Griffiths)}$$

and

$$\Lambda^\mu_{\nu'} \longrightarrow (\Lambda^{-1})^\mu_\nu \quad \text{(Griffiths)}$$

8 The Lorentz Group

A group G is a set of elements and an operation $\cdot$ with the properties:

- **(A)** closure: if $a, b \in G$ then $a \cdot b \in G$
- **(B)** associative: if $a, b, c \in G$ then $(ab)c = a(bc)$
- **(C)** identity: there exists e such that $ea = ae = a$
- **(D)** inverses: for each $a \in G$ there exists a^{-1} such that $a^{-1}a = aa^{-1} = e$

In this section, we'll show that the set of transformations between inertial references frames in special relativity form a group, which we call the Lorentz group.

Exercise R.37: (*) For $a, b \in G$, show that if $ab = e$ then $b = a^{-1}$. □

Exercise R.38: For $a, b \in G$, show that if $ab = a$ then $b = e$. □

Exercise R.39: For $a \in G$ show that $(a^{-1})^{-1} = a$. □

Valid Coorinate Transformations: The requirement that the space-time interval is invariant significantly constrains the valid coordinate transformations between inertial frames. For any four-vector v we have:

$$(v|v) = v^\mu v^\nu g_{\mu\nu}$$

but also

$$\begin{aligned}(v|v) &= v^{\alpha'} v^{\beta'} g_{\alpha'\beta'} \\ &= (v^{\mu}\Lambda^{\alpha'}_{\mu})(v^{\nu}\Lambda^{\beta'}_{\nu}) g_{\alpha'\beta'} \\ &= v^{\mu} v^{\nu} \Lambda^{\alpha'}_{\mu} g_{\alpha'\beta'} \Lambda^{\beta'}_{\nu}\end{aligned}$$

so, as v is arbitrary, we must have:

$$\Lambda^{\alpha'}_{\mu} g_{\alpha'\beta'} \Lambda^{\beta'}_{\nu} = g_{\mu\nu} \tag{14}$$

This is simply the transformation rule for any (2,0)-tensor. However, the components of the metric tensor are frame invariant, which led us to Equation 12:

$$g_{\alpha'\beta'} = \delta^{\alpha}_{\alpha'} \delta^{\beta}_{\beta'} g_{\alpha\beta}$$

We have therefore:

$$\delta^{\alpha'}_{\alpha} \delta^{\beta'}_{\beta} \Lambda^{\alpha'}_{\mu} g_{\alpha\beta} \Lambda^{\beta'}_{\nu} = g_{\mu\nu} \tag{15}$$

In the previous section, we saw that we can write $\Lambda^{\beta'}_{\nu}$ as the components of a matrix Λ. We can write this result in matrix notation as:

$$\Lambda^T G \Lambda = G \tag{16}$$

where G is the frame invariant matrix representation of the metric tensor. The need to transpose (via the symbol T) in the first factor follows from the convention we choose for Λ, that the upper index refers to the row of the matrix, in order to match the calculation in Equation 15. The δ factors are just identity matrices, which have been omitted. It is only the fact that the metric tensor has the same components in any basis that leads to this important result.

The set of all matrices Λ that satisfy Equation 16, with ordinary matrix multiplication defines the Lorentz group. Group property (B) follows directly from the associative property of matrix multiplication. The identity matrix satisfies Equation 16 so property (C) holds.

Exercise R.40: (*) Show that property (A) holds for the Lorentz group. That is, if

$$A^T G A = G$$

and

$$B^T G B = G$$

then

$$(AB)^T G (AB) = G. \qquad \square$$

Exercise R.41: (*) Show that property (D) holds for the Lorentz group. That is, if

$$A^T G A = G$$

and

$$(A^{-}1)^T G A^{-}1 = G$$

then

$$(AB)^T G(AB) = G. \qquad \square$$

Exercise R.42: (*) Show that a rotation by angle θ about the z-axis

$$R = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

is a member of the Lorentz Group. $\square$

Exercise R.43: (*) Show that a Lorentz boost along the x-axis:

$$L = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

with

$$\gamma \equiv \frac{1}{\sqrt{1-\beta^2}}$$

is a member of the Lorentz Group. $\square$

Exercise R.44: (*) Show that time reversal:

$$T = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

is a member of the Lorentz Group. $\square$

Exercise R.45: (*) Show that parity:

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

is a member of the Lorentz Group. $\square$

Exercise R.46: Show that:

$$\begin{aligned} G\,R\,G &= R^{-1} \\ G\,L\,G &= L^{-1} \\ G\,P\,G &= P^{-1} \\ G\,T\,G &= T^{-1} \end{aligned}$$

Note that this follows from the group property:

$$\begin{aligned} g_{\alpha\beta} &= g_{u'w'}\Lambda^{u'}_{\alpha}\Lambda^{w'}_{\beta} \\ g^{\gamma\alpha}g_{\alpha\beta} &= g^{\gamma\alpha}\Lambda^{u'}_{\alpha}g_{\mu'w'}\Lambda^{w'}_{\beta} \\ \delta^{\gamma}_{\beta} &= (g^{\gamma\alpha}\Lambda^{u'}_{\alpha}g_{\mu'w'})\Lambda^{w'}_{\beta} \end{aligned}$$

Notice also how nicely our notation (with primed indices) works:

$$g^{\gamma\alpha}\Lambda^{u'}_{\alpha}g_{\mu'w'} = \Lambda^{\gamma}_{w'}$$

9 Relativistic Kinematics

The principle of conservation of momentum, using the classical definition of momentum $\vec{p} = m\vec{v}$, is not consistent with special relativity. Experimental evidence supports special relativity. So we must abandon our classical notion of momentum.

If we can find a four-vector quantity related to momentum, and likewise conserved, so e.g.:

$$p^{\mu}_{A} + p^{\mu}_{B} = p^{\mu}_{C} + p^{\mu}_{D}$$

then we can simply apply a Lorentz transformation to both sides:

$$\Lambda^{\nu'}_{\mu}p^{\mu}_{A} + \Lambda^{\nu'}_{\mu}p^{\mu}_{B} = \Lambda^{\nu'}_{\mu}p^{\mu}_{C} + \Lambda^{\nu'}_{\mu}p^{\mu}_{D}$$

and obtain:

$$p^{\nu'}_{A} + p^{\nu'}_{B} = p^{\nu'}_{C} + p^{\nu'}_{D}$$

which shows that the law will hold in any frame F'. Finding this quantity is the goal of this section.

Consider two time-like events A and B, which (by definition) have a space-time interval $\Delta S^2 > 0$. We define the proper-time (τ) between these two events as:

$$\tau \equiv \sqrt{\Delta S^2}$$

The proper-time τ is equivalent to the elapsed (positive) time in the **particular** frame where the events occur at the same location. Such a frame exists because $\Delta S^2 > 0$, and calculating in that frame (call it F') we have:

$$0 < \Delta S^2 = (\Delta t')^2 - (\Delta x')^2 - (\Delta y')^2 - (\Delta z')^2$$

but in this paritcular frame we have:

$$(\Delta x')^2 = (\Delta y')^2 = (\Delta z')^2 = 0$$

so:

$$\Delta S^2 = (\Delta t')^2$$

and we see that

$$\tau \equiv \Delta t'$$

Let β be the relative speed between frame F and the frame F' in which the events A and B occur at the same location. Then we can calculate the elapsed time t in frame F according to the inverse Lorentz transformation:

$$\begin{pmatrix} t \\ x \end{pmatrix} = \begin{pmatrix} \gamma & \gamma\beta \\ \gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} \tau \\ 0 \end{pmatrix} = \begin{pmatrix} \gamma\tau \\ \gamma\tau\beta \end{pmatrix}$$

From which we see that:

$$\tau = t/\gamma \tag{17}$$

but take care that γ is for the particular frame where the events occur at the same location, as measured from the frame in which t is measured.

We use the proper-time τ to define the four-velocity η (also called proper velocity in Griffiths) by its components:

$$\eta^\mu \equiv \frac{dx^\mu}{d\tau} \tag{18}$$

Since τ and so $d\tau$ is an invariant, we see that the components η^μ transform as for a four-vector, and so, by definition, the four-velocity is a four-vector. Let's work out the coordinates in frame F by using $d\tau = dt/\gamma$:

$$\eta^\mu = \gamma \frac{dx^\mu}{dt}$$

and so for the time component:

$$\eta^0 = \gamma \frac{dt}{dt} = \gamma$$

and for the x component:

$$\eta^1 = \gamma \frac{dx}{dt} = \gamma\beta_x$$

and similar for the y and z components. So writing η as a column vector in the standard basis $\mathcal{B}$:

$$[\eta]_{\mathcal{B}} = \gamma \begin{pmatrix} 1 \\ \beta_x \\ \beta_y \\ \beta_z \end{pmatrix}$$

It is left as an exercise to show that the invariant associated with η is

$$\eta^\mu \eta_\mu = \eta^\mu \eta^\nu g_{\mu\nu} = 1 = (c^2) \tag{19}$$

Exercise R.47: Show by an explicit calculation that Equation 19 is correct. □

Next we define the momenum four-vector p by its components:

$$p^\mu \equiv m\eta^\mu \tag{20}$$

assuming that the mass of a particle is invariant, then p^μ transforms as a four-vector, and so is itself a four-vector, with

$$[p]_{\mathcal{B}} = \begin{pmatrix} \gamma m \\ \gamma m \beta_x \\ \gamma m \beta_y \\ \gamma m \beta_z \end{pmatrix}$$

In the non-relativistic limit, we can approximate γ by its Taylor series near $\beta = 0$

$$\gamma = \frac{1}{\sqrt{1-\beta^2}} = 1 + \frac{1}{2}\beta^2$$

In this limit, we see that

$$\begin{aligned} p^1 &\approx m\beta_x \\ p^2 &\approx m\beta_y \\ p^3 &\approx m\beta_z \end{aligned}$$

so that the spatial part of p reproduces the non-relativistic momentum for small β. For the temporal component, we see that:

$$p^0 \approx m + \frac{1}{2} m\beta^2$$

The second term is just the classical kinetic energy, which identifies p^0 as the relativistic energy of the particle. The first term m is a new contribution to the energy, due to special relativity, called the rest energy. We see that massive particles, even at rest, have a non-zero energy due to their mass.

Thus we have identified exactly the quantity we sought, a four-vector p which we can write as a column vector:

$$[p]_{\mathcal{B}} = \begin{pmatrix} E \\ \vec{p} \end{pmatrix}$$

where E is relativistic energy:

$$E = \gamma m$$

and $\vec{p}$ is the relativistic momentum:

$$\vec{p} = \gamma m \vec{v}$$

A common mistake by students new to relativistic kinematics is to forget the new factor γ when they encounter the familiar symbol $\vec{p}$. We have taken that symbol over to refer to the relativistic momentum, which reproduces the classical momentum only for small β.

Notice that the invariant associated with p is:

$$p^\mu p_\mu = m^2 \nu^\mu \nu_\mu = m^2$$

and also:

$$p^\mu p_\mu = E^2 - \vec{p} \cdot \vec{p}$$

so we have:

$$E^2 = m^2 + |\vec{p}|^2 \tag{21}$$

We often write this simply as:

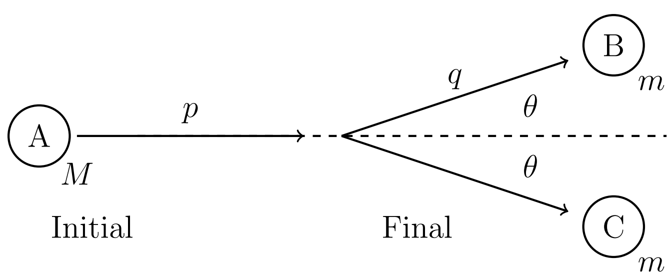
$$E^2 = m^2 + p^2 \tag{22}$$

where p is implicitly $|\vec{p}|$ and not the four-vector p. **You must memorize this equation** (or its derivation). It will not be given to you on exams, and you will need it.

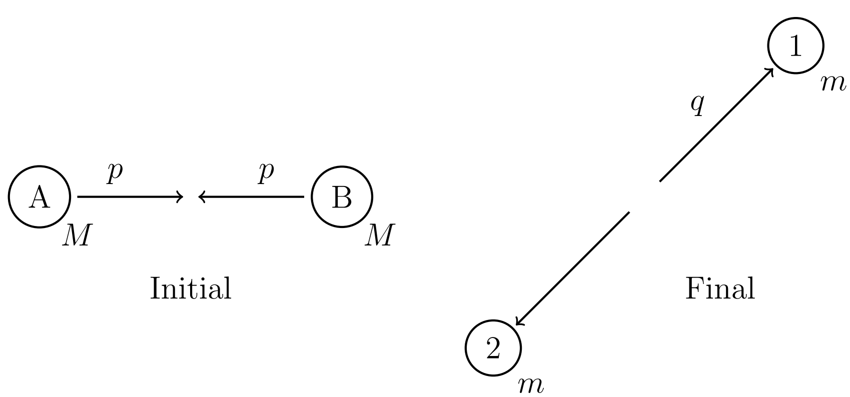
The experimental evidence confirms that the momentum four vector p is indeed conserved. In classical mechanics, we had conservation of energy and momentum separately. In relativistic kinematics, four-momentum is conserved which implies conservation of relativistic energy and relativistic momentum.

Exercise R.48: Suppose a particle of mass $m > 0$ has total energy E. Show that the speed of the particle is less than c. Because of convervation of energy, this shows that a massive particle cannot be accelerated to the speed of light by consuming a finite amount of energy. Assuming any observers would have $m > 0$, this is one explanation for why no inertial frames have a speed that exceeds the speed of light. □

Exercise R.49: Particle A of mass M decays into particles B and C, each of mass m, with $m < M/2$. Particle A has momentum p, and the momentum of particles B and C are at the same angle with respect to the momentum of Particle A. What is the magnitude of the momentum of particle B (q)? What is the angle (*theta*) between the momentum of Particle A and the momentum of Particle B? □



Exercise R.50: Particles A and B each of mass M collide to produce particles 1 and 2 each of mass m. In the center of mass frame, particles A and B have magnitude of momentum p. What is the magnitude of the momentum of particle 1 (q)?



10 Massless Particles

Special relativity provides a mechanism for massless particles to exist. Notice that:

$$\vec{p} = \frac{m\vec{v}}{\sqrt{1-\beta^2}}$$

and

$$E = \frac{m}{\sqrt{1-\beta^2}}$$

both go to zero for $m = 0$ **unless** $v = c$ in which case they are indeterminate. We still have, however,

$$E^2 = m^2 + |\vec{p}|^2$$

which yields:

$$E = |\vec{p}|$$

So special relativity allows for massless particles that travel exclusively at the speed of light. Massive particles have their momentum and energy defined by their velocity, but that is not possible for massless particles which travel at the speed of light. Special relativity has no prescription for how the energy of massless particle is determined, but quantum mechanics does. We have photons with energy given by:

$$E = \hbar\omega$$

Exercise R.51: (*) Show that a massless photon (γ) cannot decay to two massive particles (e.g. $\gamma \to e^- e^+$) in a vacuum. $\square$

11 Additional Exercises

Exercise R.52: (*) Suppose a particle of mass M has momentum p and energy E in the lab frame. It decays to three particles, A, B, and C, each with mass $M/3$. What is β of particle A in the lab frame?

Exercise R.53: (*) For the decay $a \to 1+2$ show that in the CMS the magnitude of the momentum of particle 1 is:

$$p^* = \frac{1}{2m_a}\sqrt{\left(m_a^2 - (m_1+m_2)^2\right)\left(m_a^2 - (m_1-m_2)^2\right)}$$

Hint: When duing serious algebra, don't move around numbers like m_1. Instead, do something like:

$$m_a \to a, \quad m_1 \to b, \quad m_2 \to c$$

Treat yourself right!

Exercise R.54: (*) The anti-proton was discovered via the process:

$$p + p \to p + p + p + \bar{p}$$

Calculate the beam energy at which this process is first kinematically allowed (A) for two colliding beams of protons with equal energy, and (B) for one beam of protons striking stationary protons.

Your calculation might lead you to conclude that the antiproton discovery was made using colliding beams (as in A), but in fact, the discover was made here in California, using the bevatron in a fixed target experiment (as in B). We did not yet have the ability to focus beams which is necessary to collide two beams at a high rate of collisions.

Exercise R.55: (*) The neutrino was discovered by studying the kinetic energy of the electron produces from beta decay of radioactive nuclei. Consider the nuclear decay:

$$^3_1H \rightarrow^3_1 He + e^- + \bar{\nu}_e$$

If the neutrino is neglected from the process, this becomes a two body decay. Calculate the kinetic energy:

$$K = \sqrt{p^2 + m^2} - m$$

of the electron in the rest frame of the

$$^3_1H$$

nucleus. Use $m(3^H) = 2808.921$ MeV, $m(3^He) = 2808.391$ MeV, and $m(e) = 0.511$ MeV. Compare your single value to Fig. 1.5 in Griffiths.

Exercise R.56: For two-body scattering $a + b \to 1 + 2$, show that the Lorentz invariant flux term:

$$F \equiv 4E_aE_b(v_a + v_b)$$

is equivalent to:

$$F = 4c^3\sqrt{(p_a \cdot p_b)^2 - m_a^2m_b^2c^4}$$

Hint: Calculate F^2 and then calculate $(p_a \cdot p_b)^2$.

Exercise R.57: Show that in the lab frame where b is at rest, the Lorentz invariant flux term is:

$$F = 4c^4m_bp_a$$

A Linearity of Transformation between Inertial Frames

Suppose we have frames F and F'. For simplicity, we will consider only the time and x position, for which there is a coordinate transformation:

$$\begin{aligned} x' &= f(x,t) \\ t' &= g(x,t) \end{aligned}$$

So then:

$$\begin{aligned} dx' &= \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial t}dt \\ dt' &= \frac{\partial g}{\partial x}dx + \frac{\partial g}{\partial t}dt \end{aligned}$$

and:

$$\frac{dx'}{dt'} = \frac{\dfrac{\partial f}{\partial x}\dfrac{dx}{dt} + \dfrac{\partial f}{\partial t}}{\dfrac{\partial g}{\partial x}\dfrac{dx}{dt} + \dfrac{\partial g}{\partial t}}$$

Now consider a particle moving at constant velocity dx/dt in frame F. By the invariance of the laws of physics, this particle must also move at constant velocity dx'/dt' in frame F'. This requirement implies that the partial derivatives

$$\frac{\partial f}{\partial x}, \quad \frac{\partial f}{\partial t}, \quad \frac{\partial g}{\partial x}, \quad \frac{\partial g}{\partial t}$$

are constant along the particle'strajectory.

Because any two events in space-time can be connected using a combination of particles at rest and particles moving at constant velocities, these partial derivatives must in fact be constant throughout space-time. Therefore, the most general form of the transformation is:

$$\begin{aligned} x' &= Ax + Bt + C \\ t' &= Dx + Et + F \end{aligned}$$

If we take the origin to coincide, we have:

$$\begin{aligned} x' &= Ax + Bt \\ t' &= Dx + Et \end{aligned}$$

demonstrating that the transformations between inertial framese are linear.